Memo students Monday.

WTW158 Calculus

Class Test 4 (Mo) | 17 - 21 April 2023 | Time: 15 min

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Surname: | | Initials: | |

Student number: | | Group: | |

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Determine the infinite limit : $\lim_{x\to 3^-} \frac{\sqrt{x}}{(x-3)^3}$. Motivate your answer.

$x \to 3^- \Rightarrow x < 3 \Rightarrow x - 3 < 0$

$\Rightarrow \lim_{x\to 3^-} \frac{\sqrt{x}}{(x-3)^3} = -\infty$

QUESTION 2 [3 marks]

Prove that $\lim_{x\to 0} \left(x^4 \cos \frac{2}{x}\right) = 0$.

$-1 \leq \cos\frac{2}{x} \leq 1 \quad \forall x \in \mathbb{R}\setminus\{0\}$

$\Rightarrow \quad -x^4 \leq \frac{1}{x^4}\cos\frac{2}{x} \leq x^4$

Since $\lim_{x\to 0} -x^4 = 0$

and $\lim_{x\to 0} x^4 = 0$,

it follows by the Squeeze Theorem that

$\lim_{x\to 0}\left(x^4 \cos\frac{2}{x}\right) = 0.$

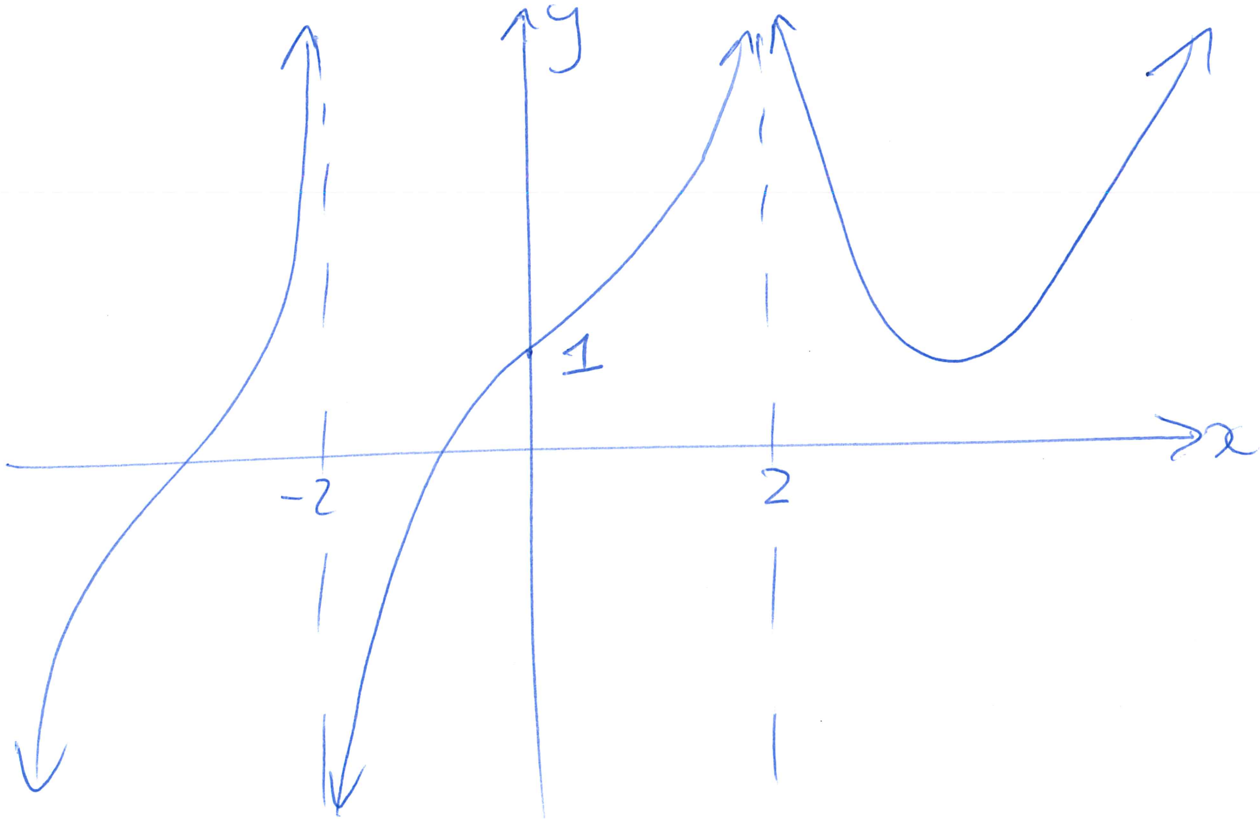
QUESTION 3 [1 mark]

Explain why $f(x) = \dfrac{x^2 - 9}{x - 3}$ is discontinuous at $a = 3$.

f(3) is not defined,
therefore f(x) is discontinuous at 3.

QUESTION 4 [3 marks]

Sketch the graph of an example of a function f that satisfies all of the given conditions:
$\lim_{x \to -\infty} f(x) = -\infty$, $\lim_{x \to \infty} f(x) = \infty$, $\lim_{x \to -2^-} f(x) = \infty$, $\lim_{x \to -2^+} f(x) = -\infty$, $\lim_{x \to 2} f(x) = \infty$ and $f(0) = 1$.



Memo students:

WTW158 Calculus

Class Test 4 (Th) 17 - 21 April 2023 Time: 15 min

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QUESTION 1 [1 mark]

Determine the infinite limit: $\lim_{x\to\frac{\pi}{2}^+} \dfrac{\sec x}{x}$.

$$\lim_{x\to\pi/2^+} \frac{\sec x}{x} = \lim_{x\to\pi/2^+} \frac{1}{x\cos x} \quad \left(\frac{1}{0}\right) \quad x\to \pi/2^+ \Rightarrow \cos x < 0$$

$$= -\infty$$

QUESTION 2 [2 marks]

Find the limit, if it exists: $\lim_{x\to 0^-} \left(\dfrac{1}{x} - \dfrac{1}{|x|}\right)$.

$$|x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$$

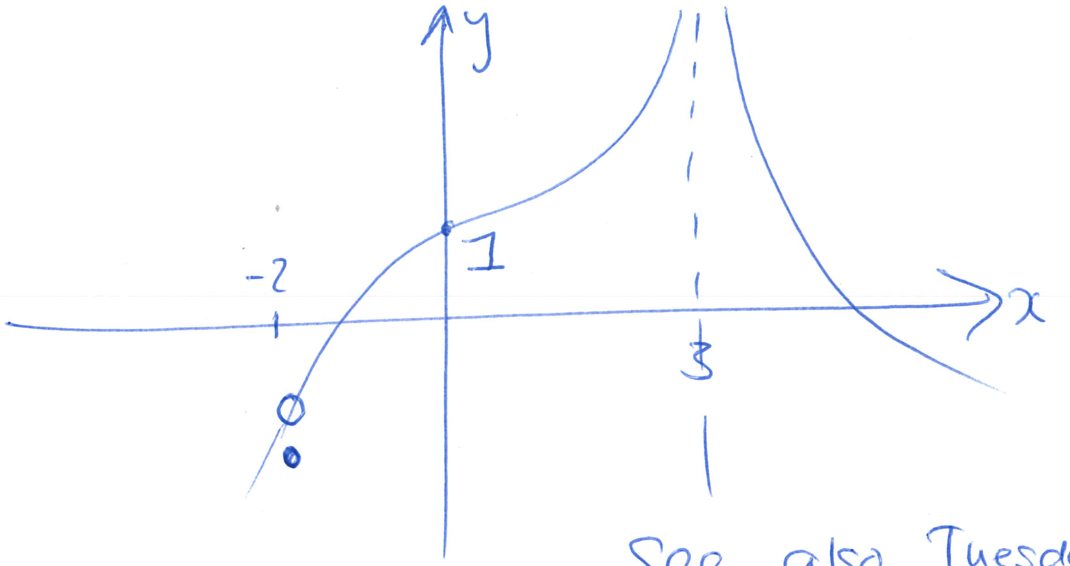
$$\Rightarrow \frac{1}{x} - \frac{1}{|x|} = \begin{cases} \frac{1}{x} - \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{x} + \frac{1}{x} & \text{if } x < 0 \end{cases}$$

$$\Rightarrow \lim_{x\to 0^-} \left(\frac{1}{x} - \frac{1}{|x|}\right) = \lim_{x\to 0^-} \frac{2}{x}$$

$$= -\infty$$

Q3 [2 marks]

Sketch the graph of a function f that is defined on $\mathbb{R}$ and continuous, except for a removable discontinuity at -2 and an infinite discontinuity at 3, $f(0) = 1$ and $f(3) = -1$.



See also Tuesday test.

QUESTION 4 [3 marks]

Find the horizontal and vertical asymptotes of the curve : $y = \dfrac{x^3 - x}{x^2 - 6x + 5}$.

$$y = \frac{x(x^2-1)}{(x-1)(x-5)} = \frac{x(x-1)(x+1)}{(x-1)(x-5)} = \frac{x(x+1)}{x-5}, \quad x \neq 1.$$

Horizontal asymptotes:

$$\lim_{x\to\infty} \frac{x^3-x}{x^2-6x+5} = \lim_{x\to\infty} \frac{x - x/x^2}{1 - 6/x + 5/x^2} = \infty$$

$$\lim_{x\to-\infty} \frac{x^3-x}{x^2-6x+5} = \lim_{x\to-\infty} \frac{x - 1/x}{1 - 6/x + 5/x^2} = -\infty$$

$\Rightarrow$ No horizontal asymptotes.

Vertical asymptotes: Test for $x=5$

$$\lim_{x\to5^+} \frac{x^3-x}{x^2-6x+5} = \lim_{x\to5^+} \frac{x(x+1)}{x-5} \left(\frac{30}{0^+}\right) = +\infty$$

$\Rightarrow$ $x=5$ is a vertical asymptote.

OR: Can determine

$\lim_{x\to5^-} \dfrac{x^3-x}{x^2-6x+5}$, which is at $-\infty$ $\Rightarrow$ $x=5$ is a vertical asymptote.

Students.

WTW158 Calculus

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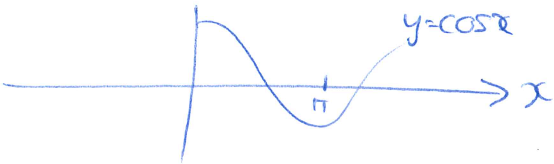
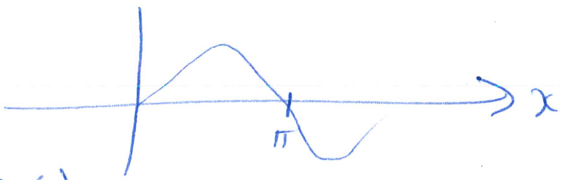
QUESTION 1 [1 mark]

Determine the infinite limit:

$\lim_{x \to \pi^-} x\cot(x)$.



$\lim_{x \to \pi^-} x \cot x$



$= \lim_{x \to \pi^-} \frac{x\cos x}{\sin x}$ $\left(\frac{\text{negative constant}}{\text{small post nr}}\right)$

$= -\infty$

QUESTION 2 [2 marks]

Find the limit: $\lim_{x \to 0} \frac{\sin 4x \ \sin 6x}{x^2}$.

$\lim_{x \to 0} \frac{\sin 4x \sin 6x}{x^2}$

$= \lim_{x \to 0} \frac{\sin 4x}{x} \times \lim_{x \to 0} \frac{\sin 6x}{x}$

$= \lim_{x \to 0} \left(\frac{\sin 4x}{x}\right)\left(\frac{4}{4}\right) \times \lim_{x \to 0} \left(\frac{\sin 6x}{x}\right)\frac{6}{6}$

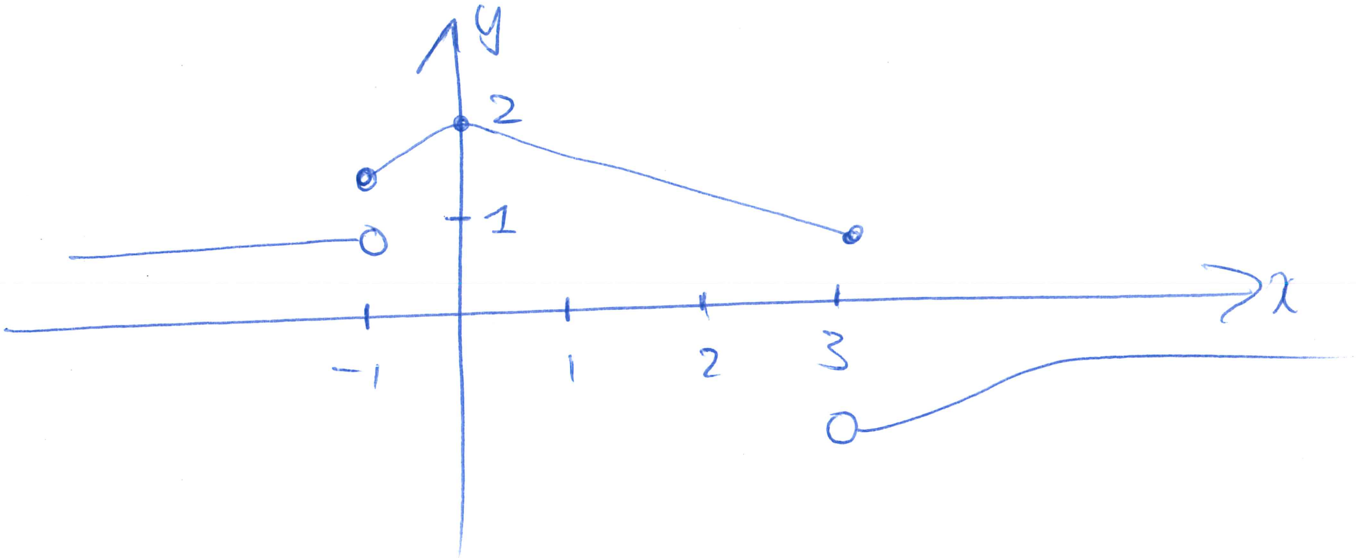
$= 4 \lim_{x \to 0} \frac{\sin 4x}{4x} \times 6 \lim_{x \to 0} \frac{\sin 6x}{6x}$

$= 4 \times 1 \times 6 \times 1$

$= 24.$

Q3: [2 marks]

Sketch the graph of a function f that is defined on $\mathbb{R}$ and continuous, except for discontinuities at -1 and 3, but continuous from the right at -1 and continuous from the left at 3, and $f(0) = 2$.



QUESTION 4 [3 marks]

Find the horizontal and vertical asymptotes of the curve : $y = \dfrac{2e^x}{e^x - 5} = f(x)$

Horizontal asymptotes

$$\lim_{x\to\infty} \frac{2e^x}{e^x-5} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{e^x(2)}{e^x(1-5/e^x)}$$

$$= \lim_{x\to\infty} \frac{2}{1-5/e^x}$$

$$= 2$$

$$\lim_{x\to-\infty} \frac{2e^x}{e^x-5} \quad \left(\frac{0}{-5}\right)$$

$$= 0$$

$\Rightarrow$ $y = 0$ and $y = 2$ are horizontal asymptotes.

Vertical asymptote:

Possible vertical asymptote where $e^x - 5 = 0$

$\Rightarrow x = \ln 5$.

[ALWAYS test for a vertical asymptote!!] MS

$\therefore$ Find $\lim_{x\to(\ln 5)^-} f(x)$

OR $\lim_{x\to(\ln 5)^+} f(x)$

only 1 necessary

$$\lim_{x\to(\ln 5)^-} \frac{2e^x}{e^x-5} = -\infty$$

$x \to (\ln 5)^-$
$\Rightarrow x < \ln 5$
$\Rightarrow e^x < 5$
$\Rightarrow e^x - 5 < 0$

OR

$$\lim_{x\to(\ln 5)^+} \frac{2e^x}{e^x-5} = +\infty$$

$\Rightarrow x = \ln 5$ is a vertical asymptote